Sarah M. Coleman, PhD

Austin, TX | **Open to relocation nationwide.**
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Summary

Bioprocess development engineer specializing in **upstream process development** and **fed-batch fermentation**, with upstream PD experience at **Bristol Myers Squibb** (biologics, CHO **cell culture**) and **Takeda** (vaccines, Vero cells). Background spans microbial strain and fermentation process development, downstream protein recovery, multi-omics analysis, and statistical modeling. PhD in Chemical Engineering and MS in Statistics, with experience in **scale-up**, **DoE**, and statistical analysis of experimental biology data in Python and R.

Work Experience

BioReNuva, Austin, TX
Bioprocess Development Engineer | Jun 2026 – Present

- Established fed-batch fermentation capability for glycolipid production with *Pseudomonas* as subject-matter expert
- Designed and implemented the company's first fed-batch bioreactor process from scratch, including equipment selection, feed strategy, control parameters, and run protocols
- Defined key performance indicators (KPIs) and identified process bottlenecks, informing the scale-up roadmap
- Authored the bioreactor standard operating procedure (SOP) and batch records; trained team members on the process

The University of Texas at Austin, Austin, TX
Postdoctoral Research Fellow (Statistics and Data Sciences) | Oct 2024 – Sep 2026

- Developed epidemiological and statistical models of environmental-health data in Dr. Roger D. Peng's group
- Wrote first-author papers in *Environmental Research Communications*, *Environmental Health* (under review), and *CHANCE* (in press), the last from a first-place JSM Data Challenge Expo entry
- Advised medical faculty on statistical methodology

The University of Texas at Austin, Austin, TX
Lecturer (Statistics and Data Sciences) | Jan 2026 – Apr 2026

- Taught a 100-student upper-division data science and machine learning course as the instructor of record, rated 4.32/5 by students, above department, college, and university averages; managed 3 TAs

The University of Texas at Austin, Austin, TX
Graduate Research Fellow (Chemical Engineering) | Aug 2019 – Aug 2024

- Designed and ran batch *Yarrowia lipolytica* fermentations in 3 L bioreactors at up to 50% (v/v) waste cooking oil, among the highest loadings reported, with CFD modeling of scale-up to 4,000 L, published in *Biotechnology Journal*
- Developed media formulations, prepared inoculum, ran reactor sterilization and probe calibration, and tracked runs by optical density, dry cell weight, and HPLC for sugars and organic acids
- Designed factorial (DoE) screens of nitrogen source, nitrogen concentration, and substrate loading to optimize media for a yeast alkane-assimilation panel, analyzed by three-way ANOVA with post-hoc correction
- Increased *Y. lipolytica* tolerance to hydrothermal liquefaction wastewater from 10% to 30% (v/v) by adaptive laboratory evolution, yielding an isolate with nearly 3-fold higher titer of the high-value polyketide triacetic acid lactone while reducing COD (chemical oxygen demand), published in *Applied Microbiology and Biotechnology*
- Evolved *Y. lipolytica* for growth on high-density waste cooking oil (submitted to *Bioresource Technology*), characterized it by genomics, transcriptomics, and metabolomics, and engineered it to produce long-chain fatty alcohols
- Developed a genome editing system using Golden Gate assembly for targeted engineering of *Y. lipolytica*, and used this system to screen a native promoter library
- Proposed and validated a linear mixed-effects model to quantitatively evaluate the performance of biological Boolean logic gates for the first time, published in *ACS Synthetic Biology*
- Mentored 4 undergraduates and 2 junior PhD students; presented results to project sponsors and collaborators

MathWorks, Remote (Natick, MA)
MATLAB Student Ambassador | Mar 2020 – Jul 2021

- Taught technical workshops on MATLAB-based modeling and data analysis to academic researchers at UT Austin

Massachusetts Institute of Technology, Cambridge, MA
Advanced Undergraduate Researcher | Sep 2018 – Aug 2019

- Developed a method of nanophosphor size sorting in Dr. Michael S. Strano's lab to improve the capacitance of nanoparticles infiltrated into a light emitting plant

Bristol Myers Squibb, Devens, MA
Upstream Process Development Intern (Biologics) | Jun 2018 – Aug 2018
- Designed and executed metabolite-controlled fed-batch cell culture experiments in 12 parallel 5 L bioreactors to systematically evaluate monoclonal antibody production performance in CHO cell lines
- Analyzed upstream bioprocess data to identify conditions that improved yield stability and process robustness

Massachusetts Institute of Technology, Cambridge, MA
Undergraduate Researcher | Sep 2016 – Aug 2018
- Expressed recombinant protein in *E. coli* with IPTG induction, harvested by centrifugation and depth filtration, performed lysis and clarification, and purified by 6xHis affinity chromatography, assessing yield and purity by SDS-PAGE, Bradford, and UV280
- Developed anaerobic assays in Dr. Gregory Stephanopoulos's lab to characterize the carbon assimilation of *Moorella thermoacetica* enzymes in the Wood-Ljungdahl pathway

Takeda Pharmaceuticals, Cambridge, MA
Process Development Intern (Vaccines) | May 2017 – Aug 2017
- Optimized Vero cell culture to improve vaccine production in a next-generation manufacturing system
- Ran tangential flow filtration (TFF) for vaccine manufacturing, and helped design experiments to improve process scalability and production efficiency
- Performed analytical assays to determine infectious vaccine titer (CFU)

Education

The University of Texas at Austin, Austin, TX
Ph.D. in Chemical Engineering, Aug 2024 (Advisor: Dr. Hal S. Alper)
The University of Texas at Austin, Austin, TX
M.S. in Statistics, May 2024
Massachusetts Institute of Technology, Cambridge, MA
S.B. in Chemical-Biological Engineering, Jun 2019 (Research advisors: Dr. Michael S. Strano, Dr. Gregory Stephanopoulos)

Technical Skills

- **Bioprocessing & Fermentation**: Batch and fed-batch microbial fermentation, bioreactor operation with DO and pH control (BioFlo/CelliGen with BioCommand, DeltaV), shake flask culture, high-throughput growth screening (BioLector, 96-well plates), OD600, dry cell weight, off-gas analysis, scale-up, kLa and mass transfer, media and feed development, seed train, tangential flow filtration (TFF), SOPs, batch records, tech transfer documentation
- **Synthetic Biology & Metabolic Engineering**: Metabolic engineering, genome editing (Golden Gate and Gibson assembly, CRISPR), plasmid design and cloning (*E. coli*), colony selection, strain engineering, promoter library screening, adaptive laboratory evolution
- **Cell Culture & Molecular Biology**: Yeast, bacterial, and mammalian (CHO, Vero) cell culture, aseptic technique, BSL-2 and BSL-2+, cGLP, recombinant protein expression (IPTG induction), 6xHis affinity chromatography, SDS-PAGE, Vi-CELL, Cedex Bio, NucleoCounter, flow cytometry, qPCR, protein purification, Western blotting
- **Computational Biology & Data Science**: Python, R, SQL, MATLAB, JMP, statistical modeling, machine learning, data visualization, DoE, CFD (Ansys CFX)
- **Bioinformatics & Omics**: RNA-seq and Tag-seq (DESeq2), genomic variant analysis, genome annotation, metabolomics, high-performance computing, Docker
- **Analytical Chemistry**: HPLC, LC-MS/MS, GC-MS, headspace GC, UV-Vis, assay development, plate reader assays
- **Software**: Benchling, LIMS, Linux, Bash, Git/GitHub, Jupyter, LaTeX, GraphPad Prism

Select Honors and Awards

- **First Place, Professional Division at the Data Challenge Expo, JSM (Joint Statistical Meetings)**, 2025
- **National Science Foundation Graduate Research Fellowship (NSF GRFP)**, 2021
- **Provost's Graduate Excellence Fellowship, UT Austin**, 2019
- **Individual Accomplishment Citation, MIT Department of Chemical Engineering**, 2018 & 2019
  Only known undergraduate to win this award two years in a row, for volunteer work and service to the department

Publications

14 peer-reviewed publications, seven as first or co-first author. Full list: sarahmcoleman.github.io/papers.html